

Interview Practice Platform

AI Resume Matching & Feedback

Database Systems Engineering and Distributed Backend Development (25CS1302E)

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Problem Statement

Job seekers struggle to find roles that genuinely fit their skills, and even when they land an interview, most prepare with generic questions that don't reflect the role — or get zero feedback on how they actually performed.

! **No personalized job matching**
Resumes are manually screened or keyword-filtered, missing better-fit roles a candidate never sees.

! **Generic interview prep**
One-size-fits-all question banks don't target the specific role's likely questions or a candidate's own gaps.

! **No feedback loop**
Practicing alone gives no insight into answer quality, pacing, or delivery — the things that actually decide interviews.

The Gap Today

75%+

of resumes are filtered out by ATS or recruiters before a human ever reads them

Few

candidates can access affordable, role-specific interview coaching

Zero

structured feedback on delivery when practicing alone at home

Existing System

How job seekers currently handle resume screening and interview prep, and where each approach falls short.

Keyword-based ATS screening

Applicant Tracking Systems filter resumes by matching exact keywords against a job description.

Limitation: Misses qualified candidates who phrase skills differently, and gives no visibility into why a resume was rejected.

Static online question banks

Generic "top 50 interview questions" lists and PDFs used for self-prep.

Limitation: Not tailored to a specific job or the candidate's own resume gaps, so practice rarely matches the real interview.

Peer-to-peer mock platforms

Services like Pramp/Exponent pair candidates for live mock interviews.

Limitation: Depends on finding and scheduling a human partner, and feedback quality varies with the peer's own experience.

Delivery-only coaching tools

Apps such as Yoodli score pace, filler words and confidence from speech.

Limitation: Coaches how something is said, not whether the content answers a role-specific question well.

Proposed System

An AI-powered platform that matches candidates to roles from their resume, generates role-specific mock interview questions, captures spoken answers in-browser, and returns structured feedback — all running on free-tier and open-source infrastructure.



Resume & Matching Service

- spaCy — resume parsing, skill/experience extraction
- sentence-transformers — embedding-based job similarity
- Seeded job dataset for match ranking
- REST endpoints consumed by the Node backend



Interview & Feedback Engine

- getUserMedia + MediaRecorder — in-browser mic/camera capture
- faster-whisper — speech-to-text transcription
- Groq/Gemini LLM — question generation & feedback
- MongoDB — sessions, transcripts & scores

Objectives

What this project sets out to achieve, mapped to the problems identified above.

01

Accurate resume parsing

Extract skills, experience and role signals from a candidate's resume reliably.

02

Smarter job matching

Rank job fit using semantic similarity rather than keyword filters alone.

03

Role-specific mock interviews

Generate interview questions tailored to the matched job and the candidate's gaps.

04

Actionable feedback

Score spoken answers on content and delivery so candidates know what to improve.

05

Track progress over time

Store sessions, transcripts and scores so candidates can see improvement across attempts.

06

Keep it accessible

Build entirely on free-tier and open-source infrastructure so it stays affordable.

Scope

What this project covers in its current phase, and what's intentionally left for later.

In Scope

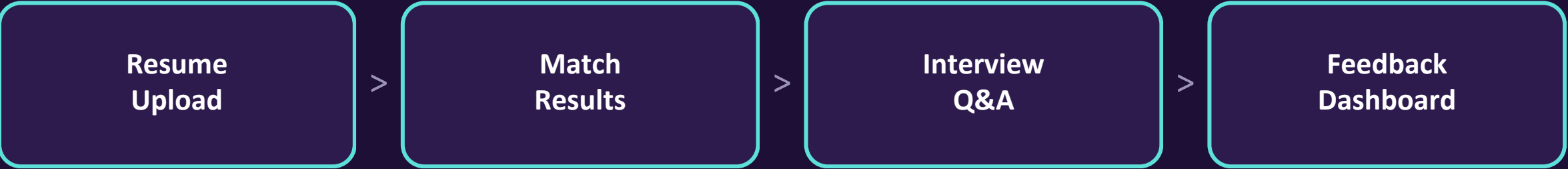
- Resume upload and NLP-based parsing of skills & experience
- Job matching against a seeded dataset using embedding similarity
- AI-generated, role-specific mock interview questions
- In-browser audio/video capture of spoken answers
- Speech-to-text transcription and LLM-based feedback scoring
- Session history and progress tracking for each candidate

Out of Scope (for now)

- Live job postings via LinkedIn/Indeed APIs (static seed dataset only)
- Body language / video analysis of the candidate
- Voice modulation (pitch, tone) scoring
- Recruiter-facing dashboard or hiring pipeline
- Multi-language interview support
- Native mobile app (web-only for this phase)

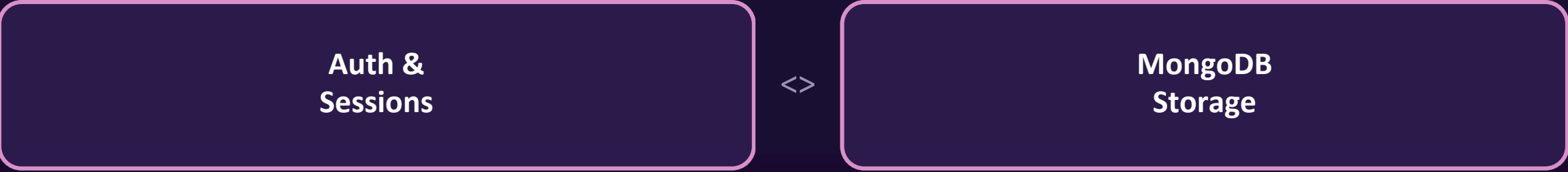
Basic Architecture

Frontend (React)



↓ REST calls, auth via Node backend

Backend (Node.js + MongoDB)



↓ Orchestrates calls to the AI microservice

AI Service (Python / Flask)



Design Notes

- Node and Python are separate services talking over REST — AI logic stays isolated and swappable.
- All AI components run on free-tier / open-source tools: Groq, faster-whisper, sentence-transformers.
- MongoDB's flexible schema suits variable-length transcripts and evolving question sets.

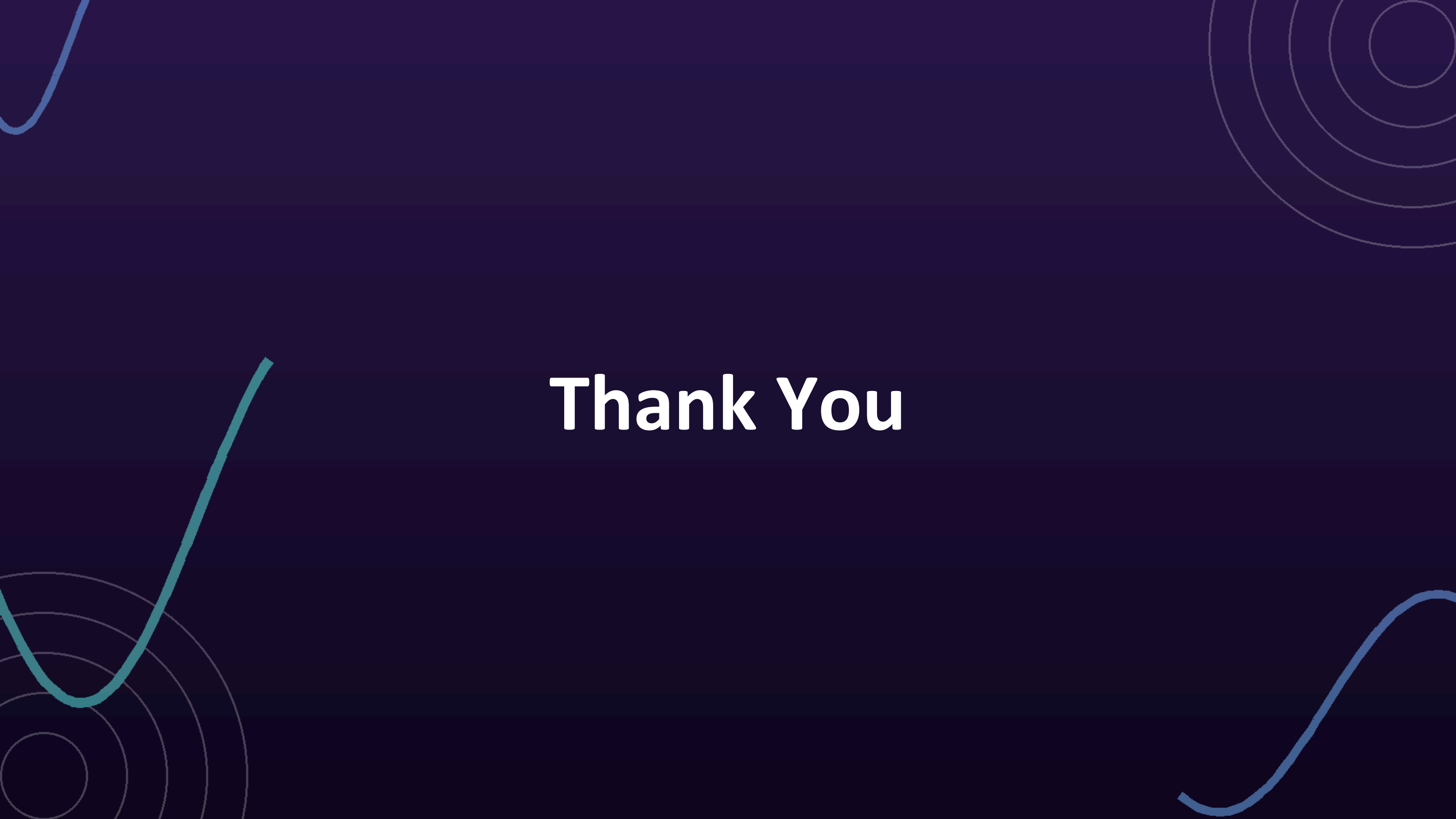
Technology Identification

Software Stack

- React.js
- Node.js + Express
- Python (Flask)
- spaCy
- sentence-transformers
- faster-whisper
- Groq / Gemini API
- MongoDB Atlas
- Git / GitHub

Hardware Requirements

- Development workstation
- Webcam & microphone
- Cloud server / MongoDB Atlas instance
- Stable internet for API calls & sync



Thank You